

Sorting Algorithms in C++

This repository contains implementations of the most commonly used **sorting algorithms** in **C++**. Each program accepts user input, sorts the array, displays the sorted output, and measures the execution time using the C++ `<chrono>` library.

Algorithms Included

- Bubble Sort
- Insertion Sort
- Selection Sort
- Merge Sort
- Quick Sort

Project Structure

```
Sorting-Algorithms/  
├─ bubblesort.cpp  
├─ insertionsort.cpp  
├─ selectionsort.cpp  
├─ mergesort.cpp  
├─ quicksort.cpp  
└─ README.md
```

Features

- Written in C++
- User input for array elements
- Displays sorted array
- Measures execution time in microseconds
- Shows the Best, Average, and Worst case time complexity for each algorithm

Time Complexity

Algorithm	Best Case	Average Case	Worst Case
Bubble Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$

Algorithm	Best Case	Average Case	Worst Case
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$

Requirements

- C++ Compiler (GCC, G++, MinGW, or MSVC)
- C++11 or later

How to Run

Compile

```
g++ bubblesort.cpp -o bubble
```

Execute

```
./bubble
```

Repeat the same steps for the other `.cpp` files.

Learning Objectives

This project helps in understanding:

- Comparison-based sorting algorithms
- Algorithm analysis
- Time complexity
- Performance measurement using `<chrono>`
- Basic C++ programming concepts

Technologies Used

- C++
- Standard Template Library (STL)
- Chrono Library

Sample Output

```
Enter number of elements:
5
```

```
Enter elements:
```

```
5 2 8 1 3
```

Sorted Array:

```
1 2 3 5 8
```

Execution Time: 12 microseconds



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★ If you found this project helpful, consider giving it a **Star** on GitHub!